Probability Mass Function and Expected Value

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Question

The random variable X takes on the values 1, 2 or 3 with probabilities $\frac{2+5P}{5}$, $\frac{1+3P}{5}$, and $\frac{1.5+2P}{5}$, respectively. The values of P and $E[X]$ are respectively:

a) $0.05, 1.87$
b) $1.90, 5.87$
c) $0.05, 1.10$
d) $0.25, 1.40$

Theory: Properties of a PMF

For any discrete random variable X, the Probability Mass Function (PMF) must satisfy two main properties:

1. The probability of each outcome must be non-negative:

$$P(X = x_i) \geq 0 \quad \text{for all } i \tag{1}$$

2. The sum of all probabilities over the entire sample space must equal 1:

$$\sum_i P(X = x_i) = 1 \tag{2}$$

We will use the second property to find the unknown variable P.

Step 1: Finding the value of P

According to the sum of probabilities rule:

$$P(X = 1) + P(X = 2) + P(X = 3) = 1 \tag{3}$$

Substituting the given expressions:

$$\frac{2 + 5P}{5} + \frac{1 + 3P}{5} + \frac{1.5 + 2P}{5} = 1 \tag{4}$$

Combine the numerators over the common denominator:

$$\frac{(2 + 1 + 1.5) + (5P + 3P + 2P)}{5} = 1 \tag{5}$$

$$\frac{4.5 + 10P}{5} = 1 \tag{6}$$

Step 1: Finding the value of P

Multiplying both sides by 5:

$$4.5 + 10P = 5 \tag{7}$$

Subtracting 4.5 from both sides:

$$10P = 5 - 4.5 \tag{8}$$

$$10P = 0.5 \tag{9}$$

Solving for P:

$$P = \frac{0.5}{10} = \mathbf{0.05} \tag{10}$$

Step 2: Deriving the PMF

Now substitute $P = 0.05$ back into the original probability expressions to find the exact PMF:

For $X = 1$:

$$P(X = 1) = \frac{2 + 5(0.05)}{5} = \frac{2 + 0.25}{5} = \frac{2.25}{5} = \mathbf{0.45} \tag{11}$$

For $X = 2$:

$$P(X = 2) = \frac{1 + 3(0.05)}{5} = \frac{1 + 0.15}{5} = \frac{1.15}{5} = \mathbf{0.23} \tag{12}$$

For $X = 3$:

$$P(X = 3) = \frac{1.5 + 2(0.05)}{5} = \frac{1.5 + 0.10}{5} = \frac{1.60}{5} = \mathbf{0.32} \tag{13}$$

Step 3: Calculating Expected Value $E[X]$

The expected value of a discrete random variable is given by:

$$E[X] = \sum_x x \cdot P(X = x) \tag{14}$$

Substitute the values from our derived PMF:

$$E[X] = (1 \cdot 0.45) + (2 \cdot 0.23) + (3 \cdot 0.32) \tag{15}$$

$$E[X] = 0.45 + 0.46 + 0.96 \tag{16}$$

$$E[X] = \mathbf{1.87} \tag{17}$$

Conclusion: The values of P and $E[X]$ are 0.05 and 1.87, respectively. This matches option **(a)**.

Step 4: Calculating the Cumulative Distribution Function (CDF)

The Cumulative Distribution Function, $F_X(x)$, represents the probability that the random variable X will take a value less than or equal to x:

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} P(X = x_i) \tag{18}$$

Using our derived PMF, we accumulate the probabilities step-by-step:

- **For** $x < 1$**:** $F_X(x) = 0$
- **For** $1 \leq x < 2$**:** $F_X(x) = P(X = 1) = \mathbf{0.45}$
- **For** $2 \leq x < 3$**:** $F_X(x) = P(X = 1) + P(X = 2) = 0.45 + 0.23 = \mathbf{0.68}$
- **For** $x \geq 3$**:** $F_X(x) = 0.68 + P(X = 3) = 0.68 + 0.32 = \mathbf{1.00}$

Step 4: CDF Piecewise Function

Summarizing the calculations, we can formally write the CDF as a piecewise step function:

$$F_X(x) = \begin{cases} 0, & \text{if } x < 1 \\ 0.45, & \text{if } 1 \leq x < 2 \\ 0.68, & \text{if } 2 \leq x < 3 \\ 1.00, & \text{if } x \geq 3 \end{cases} \tag{19}$$

This perfectly matches the expected behavior of a CDF, where it starts at 0 and monotonically increases to 1.

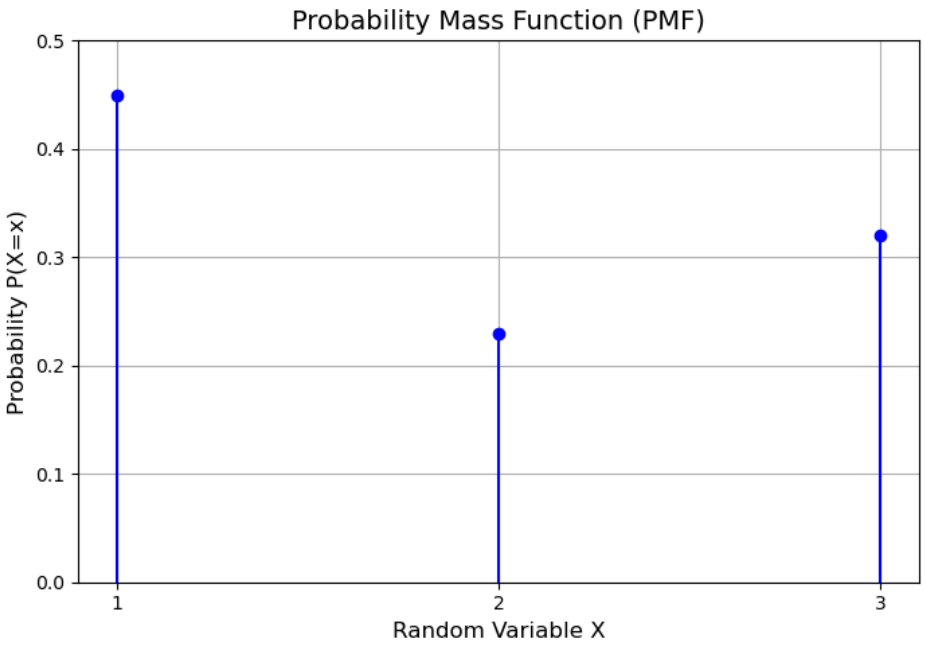
Python Code

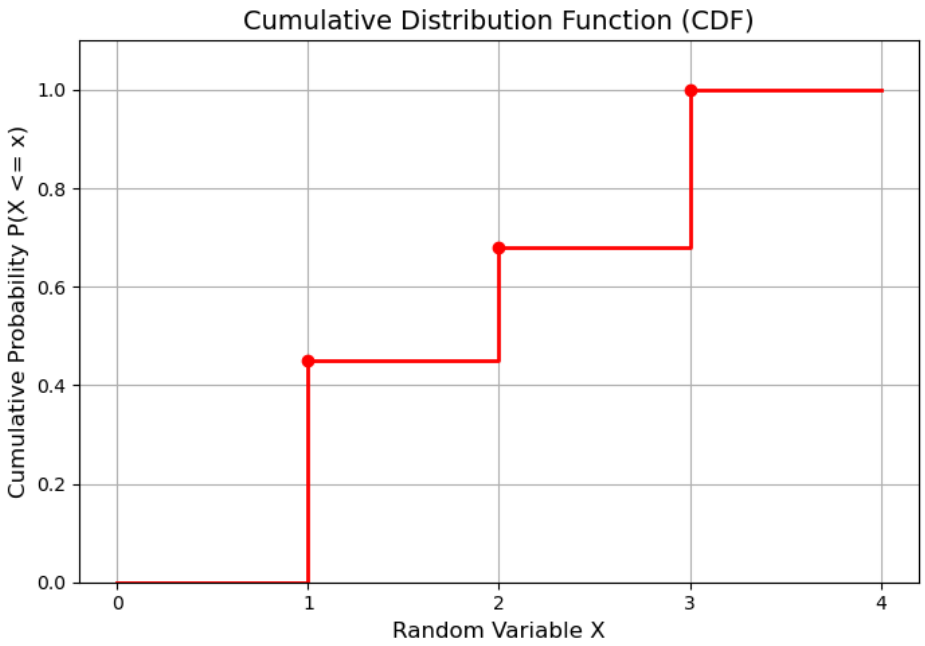
```python
import matplotlib.pyplot as plt
import numpy as np

X = np.array([1, 2, 3])
prob = np.array([0.45, 0.23, 0.32])

cdf = np.cumsum(prob)

fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(14, 5))

ax1.stem(X, prob, basefmt=" ", linefmt='b-')
ax1.set_title('Probability Mass Function (PMF)', fontsize=14)
ax1.set_xlabel('Random Variable X', fontsize=12)
ax1.set_ylabel('Probability P(X=x)', fontsize=12)
ax1.set_ylim(0, 0.5)
ax1.grid()
```

Python Code

```
x_step = [0, 1, 2, 3, 4]
y_step = [0, cdf[0], cdf[1], cdf[2], cdf[2]]

ax2.step(x_step, y_step, color='r')

ax2.plot(X, cdf)

ax2.set_title('Cumulative Distribution Function (CDF)', fontsize
    =14)
ax2.set_xlabel('Random Variable X', fontsize=12)
ax2.set_ylabel('Cumulative Probability P(X <= x)', fontsize=12)
ax2.set_ylim(0, 1.1)
ax2.grid()

plt.show()
```

Plot


Probability Mass Function (PMF)
Probability P(X=x)
Random Variable X
0.0
0.1
0.2
0.3
0.4
0.5
1
2
3


Cumulative Distribution Function (CDF)
Cumulative Probability P(X <= x)
Random Variable X
0.0
0.2
0.4
0.6
0.8
1.0
0
1
2
3
4

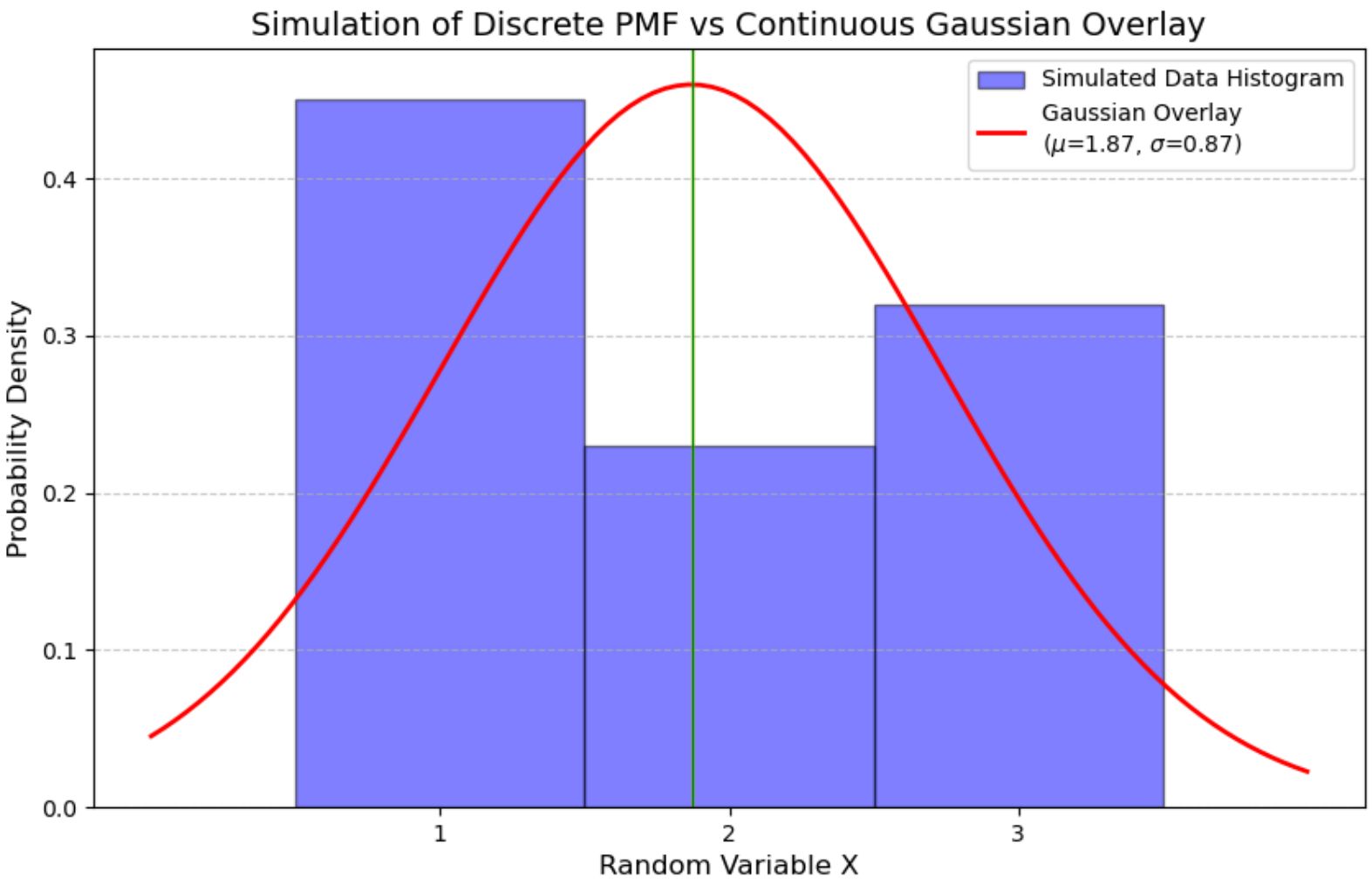
Python Code

```python
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

X = [1, 2, 3]
prob = [0.45, 0.23, 0.32]

samples = 100000
data = np.random.choice(X, size=samples, p=prob)
mu = np.mean(data)
sigma = np.std(data)

print(f"Theoretical Mean: 1.870  | Simulated Mean: {mu:.3f}")
print(f"Theoretical Std:  0.868  | Simulated Std:  {sigma:.3f}")

plt.figure()
bins = [0.5, 1.5, 2.5, 3.5]
plt.hist(data, bins=bins, density=True, alpha=0.5, color='blue',
         edgecolor='black', label='Simulated Data Histogram')
```

Python Code

```
x_axis = np.linspace(0, 4, 100)
gaussianCurve = norm.pdf(x_axis, mu, sigma)

plt.plot(x_axis, gaussianCurve, 'r-', linewidth=2,
         label=f'Gaussian Overlay\n($\mu$={mu:.2f}, $\sigma$={
    sigma:.2f})')

plt.axvline(1.87,color = "yellow",linewidth=1)
plt.axvline(mu,color = "green",linewidth=1)

plt.title('Simulation of Discrete PMF vs Continuous Gaussian
    Overlay', fontsize=14)
plt.xlabel('Random Variable X', fontsize=12)
plt.ylabel('Probability Density', fontsize=12)
plt.xticks(X)
plt.legend()
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```

Plot


Simulation of Discrete PMF vs Continuous Gaussian Overlay
Simulated Data Histogram
Gaussian Overlay
(μ=1.87, σ=0.87)
Probability Density
0.0
0.1
0.2
0.3
0.4
1
2
3
Random Variable X